DRIVE BELT AUTO TENSIONER INSPECTION [SKYACTIV-D 1.5]

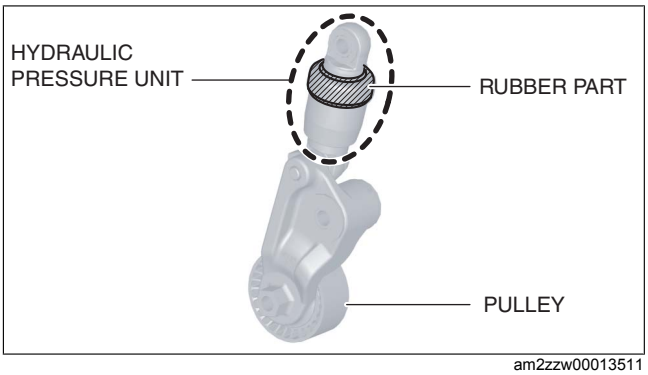
id0110q2801700

Warning

- **A hot engine can cause severe burns. Turn off the engine and wait until it is cool before servicing.**

Caution

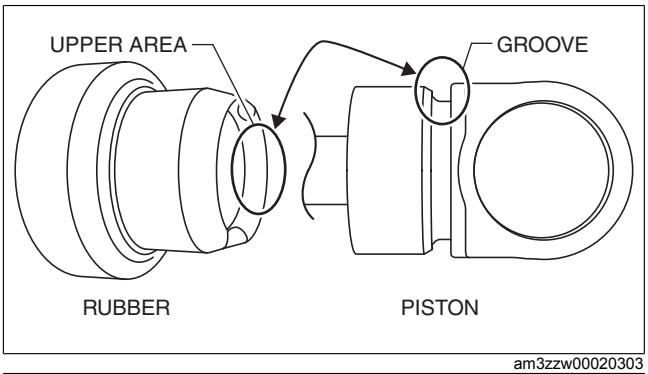
- **If the rubber part of the drive belt auto tensioner is damaged by a screwdriver or other tool, oil in the hydraulic pressure unit could leak and the tensioner may not operate normally. Be careful not to damage the rubber part of the drive belt auto tensioner.**



am2zzw00013511

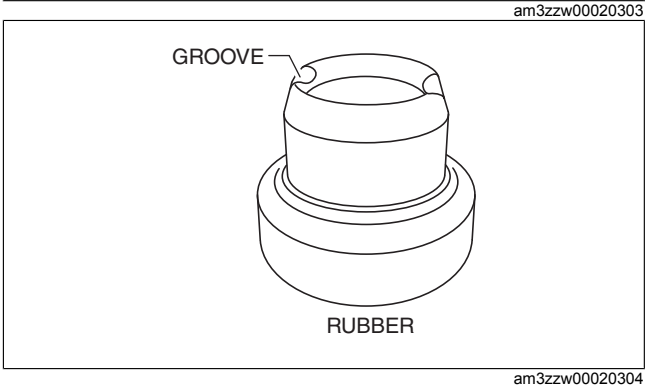
Note

- Oil is sealed in the upper area of the rubber part and the piston groove shown in the figure.



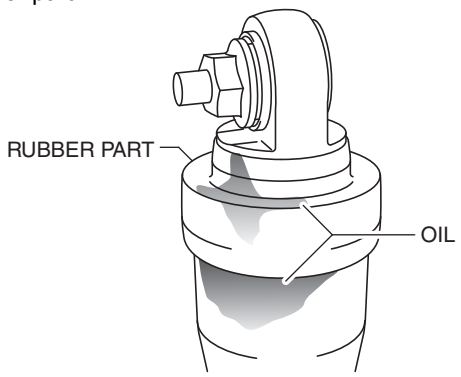
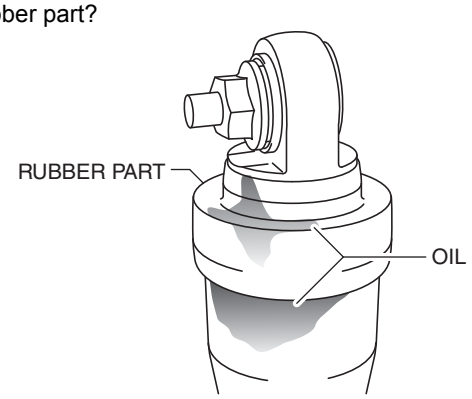
am3zzw00020303

- If the internal pressure of the hydraulic pressure unit is the specified value or more, the internal pressure will escape from the groove. At this time, oil adhering to the inner wall may seep out, however, this does not indicate a malfunction.



am3zzw00020304

Hydraulic Pressure Unit Inspection

STEP	INSPECTION		ACTION
1	**PURPOSE: INSPECT CONDITION OF RUBBER PART OF DRIVE BELT AUTO TENSIONER** • Is oil seeping out from the upper or lower side of the rubber part?  	Yes	Go to the next step.
		No	The hydraulic pressure unit is normal therefore the inspection is completed.
2	**PURPOSE: INSPECT CONDITION OF RUBBER PART OF DRIVE BELT AUTO TENSIONER** • Wipe off any dust or oil on the surface of the rubber part. • Visually inspect the rubber part for damage or separation. • Is the rubber part damaged?	Yes	Replace the drive belt auto tensioner because the rubber part is damaged and there is oil leakage. (See DRIVE BELT AUTO TENSIONER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**PURPOSE: INSPECT CONDITION OF HYDRAULIC PRESSURE UNIT OF DRIVE BELT AUTO TENSIONER** • Insert a wrench into the hexagonal area of the drive belt auto tensioner. • Apply force to the wrench and force the drive belt auto tensioner to stroke. • Does the resistance increase when the drive belt auto tensioner compresses, and decrease when it expands?	Yes	Verify that the drive belt assembly condition is normal, then go to the next step.
		No	**Resistance is high in up and down directions:** • Replace the drive belt auto tensioner due to hydraulic pressure unit damage or rotation axis damage. • (See DRIVE BELT AUTO TENSIONER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) **Resistance is low in up and down directions:** • Replace the drive belt auto tensioner due to oil leakage or functional part damage. • (See DRIVE BELT AUTO TENSIONER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4	**PURPOSE: INSPECT CONDITION OF RUBBER PART OF DRIVE BELT AUTO TENSIONER** • Is oil seeping out from the upper or lower side of the rubber part?  	Yes	**Oil is leaking from upper area of rubber part:** • The hydraulic pressure unit is normal therefore the inspection is completed. **Oil is leaking from lower area of rubber part:** • Replace the drive belt auto tensioner due to a poor seal on the rubber part. • (See DRIVE BELT AUTO TENSIONER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	The hydraulic pressure unit is normal therefore the inspection is completed.

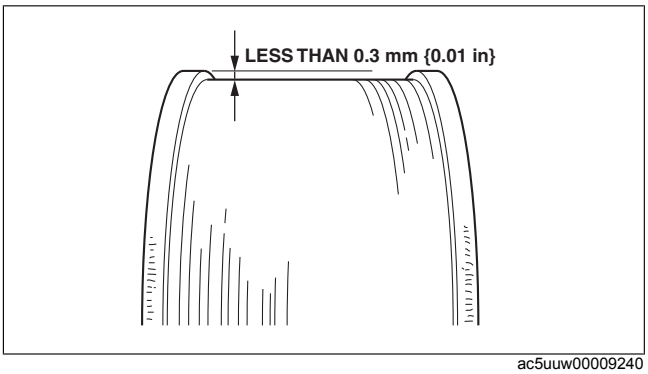
Pulley Inspection

1. Remove the generator drive belt. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
2. Rotate the drive belt auto tensioner pulley by hand and verify that it rotates smoothly.
   - If it does not rotate smoothly, replace the drive belt auto tensioner. (See DRIVE BELT AUTO TENSIONER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

3. Inspect the contact surface of the drive belt auto tensioner pulley belt and verify that the height difference is less than the maximum specification.
   - If it is the maximum specification or more, replace the drive belt auto tensioner. (See DRIVE BELT AUTO TENSIONER REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)

   Maximum height difference of pulley
   0.3 mm {0.01 in}

4. Install the generator drive belt. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



ac5uuw00009240